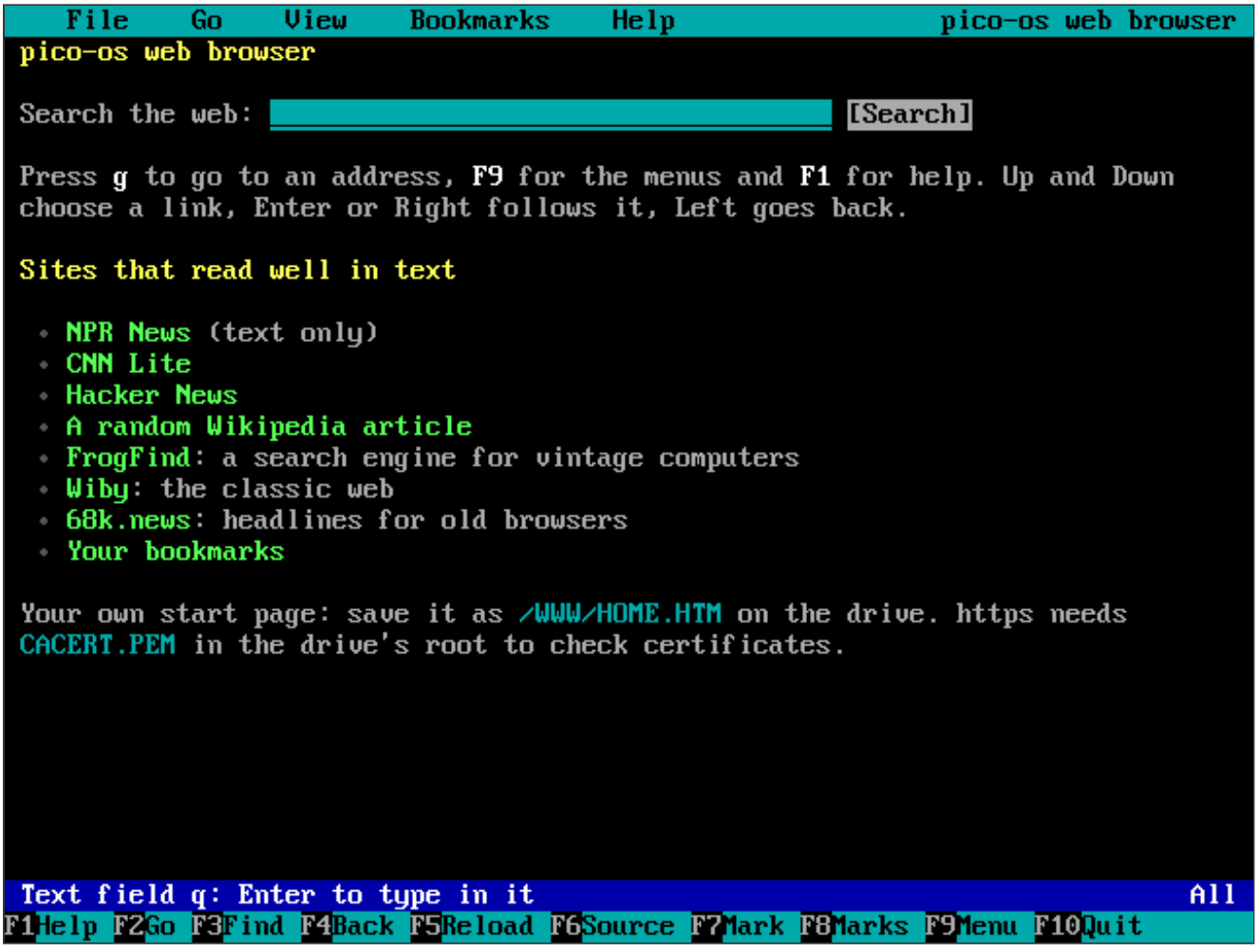
Pico 2 mini OS Web Browser

Pico 2 mini OS · October 29, 2026

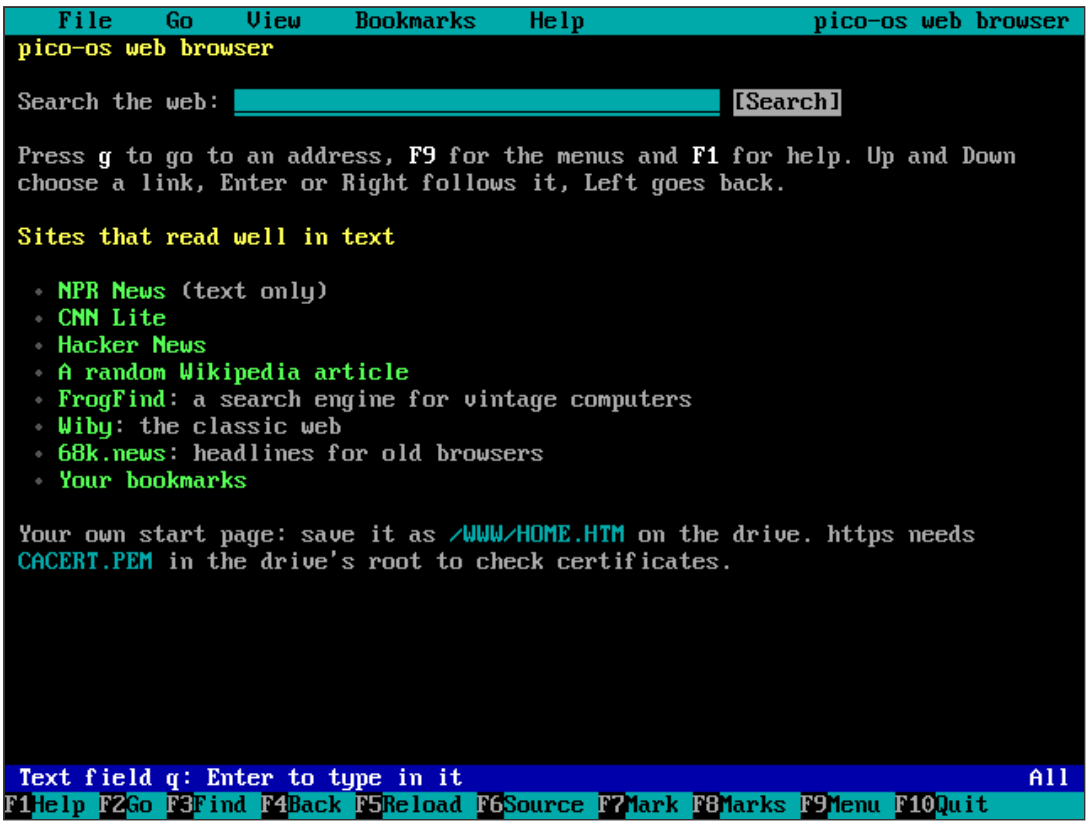


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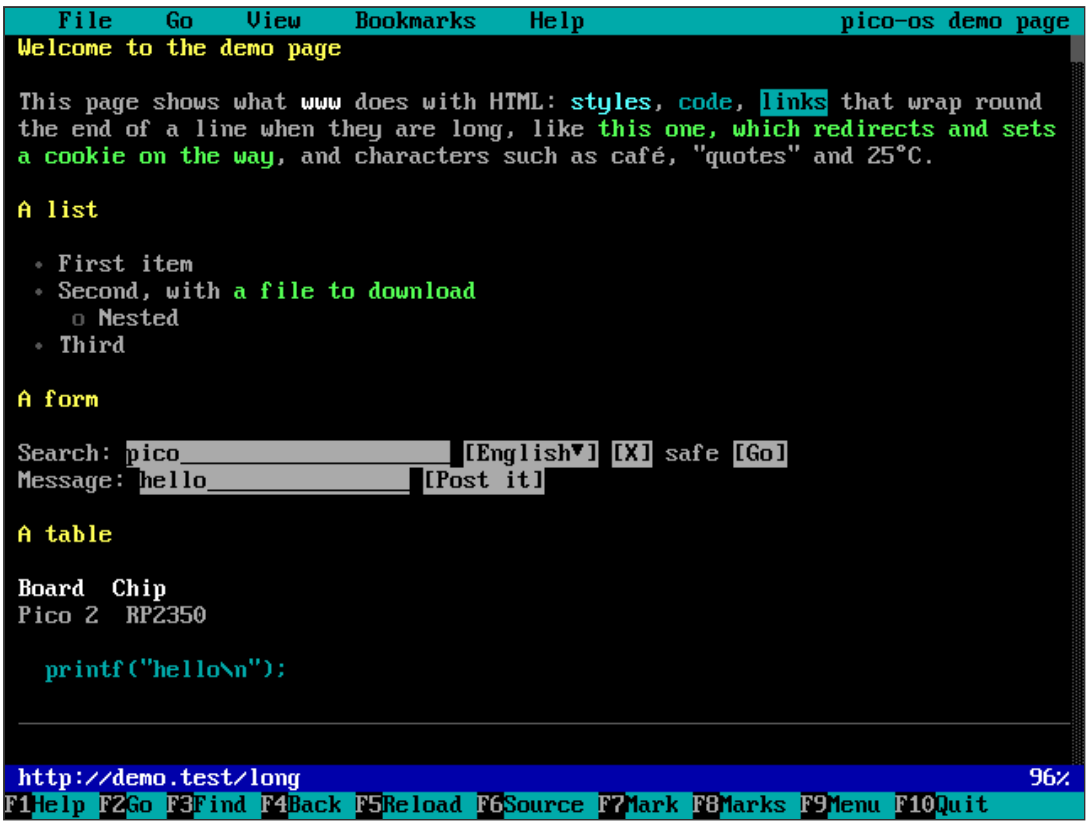
`www` (or `lynx`) is a text web browser in the manner of lynx: it loads pages over http and https (the same network and TLS code as `wget`) and shows them as text on the 80x30 screen, with their links and simple forms. There are no pictures, scripts or styles; most news, reference and search pages read well. It needs WiFi (`wifi connect`), and for https the certificate file `CACERT.PEM` in the drive's root (as for `wget`).

Starting



- `www` opens the start page: `/WWW/HOME.HTM` on the drive if there is one, else a built-in page with a search box and some sites that work well in text.
- `www text.npr.org` (or `www https://text.npr.org/`) opens that address. Without `http://` or `https://`, https is used.
- `www /DOCS/PAGE.HTM` opens a page on the drive (also `file:///...`).
- `www raspberry pi pico` searches the web for those words (DuckDuckGo's HTML page).

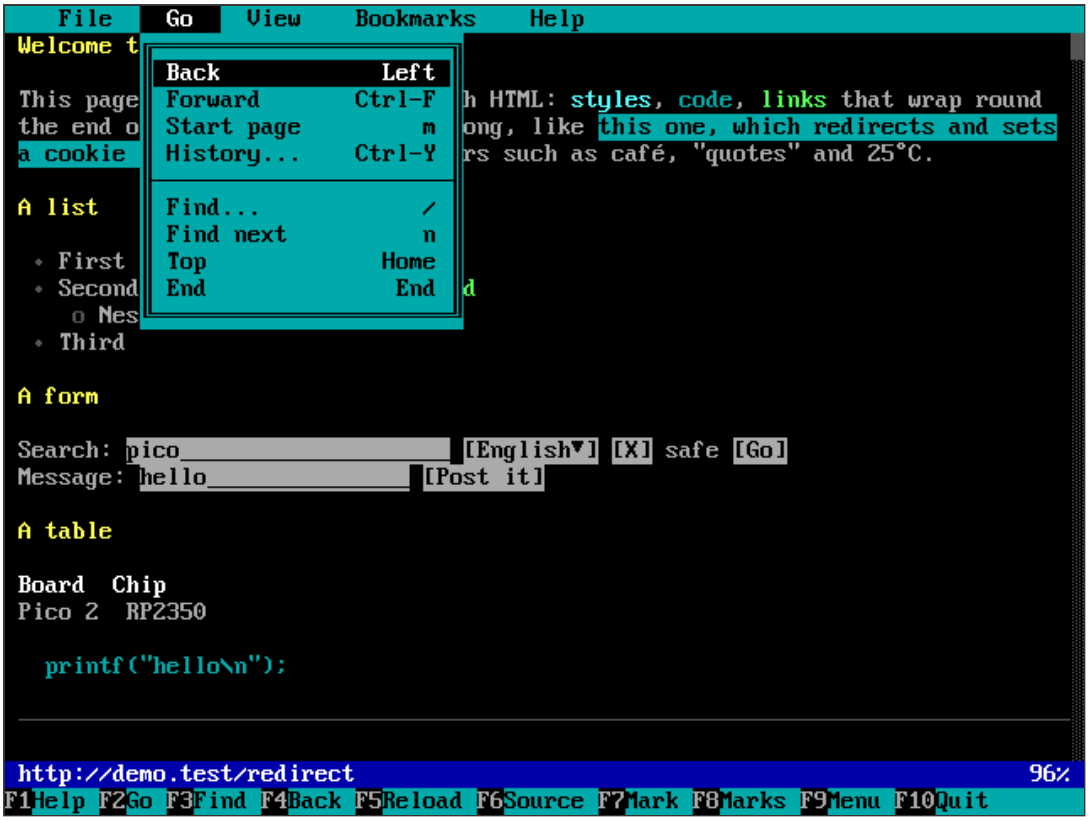
The screen



- **The menu bar** at the top: File, Go, View, Bookmarks and Help, as in mc and the tracker. F9 (or Alt and a title's first letter, or a click) opens it. The page's title is on the right.
- **The page**, 78 characters wide. Headings are yellow, bold text white, italics light cyan, code cyan, a picture's alt text magenta in `[brackets]`. **Links are green**; the link chosen is black on cyan. Form fields are grey, with `____` for the room in a text field, `[X]` for a checkbox, `(*)` for a radio button and `[choice▼]` for a list.
- **The scroll bar** on the right (when the page is longer than the screen): click it to jump there.
- **The status line** says where the chosen link goes (or what the form field is), and how far down the page is shown. While a page loads it shows the progress: looking up the name, connecting, the TLS handshake, the kilobytes so far. Esc stops loading and shows what arrived.
- **The key bar** at the bottom: click a key to press it.

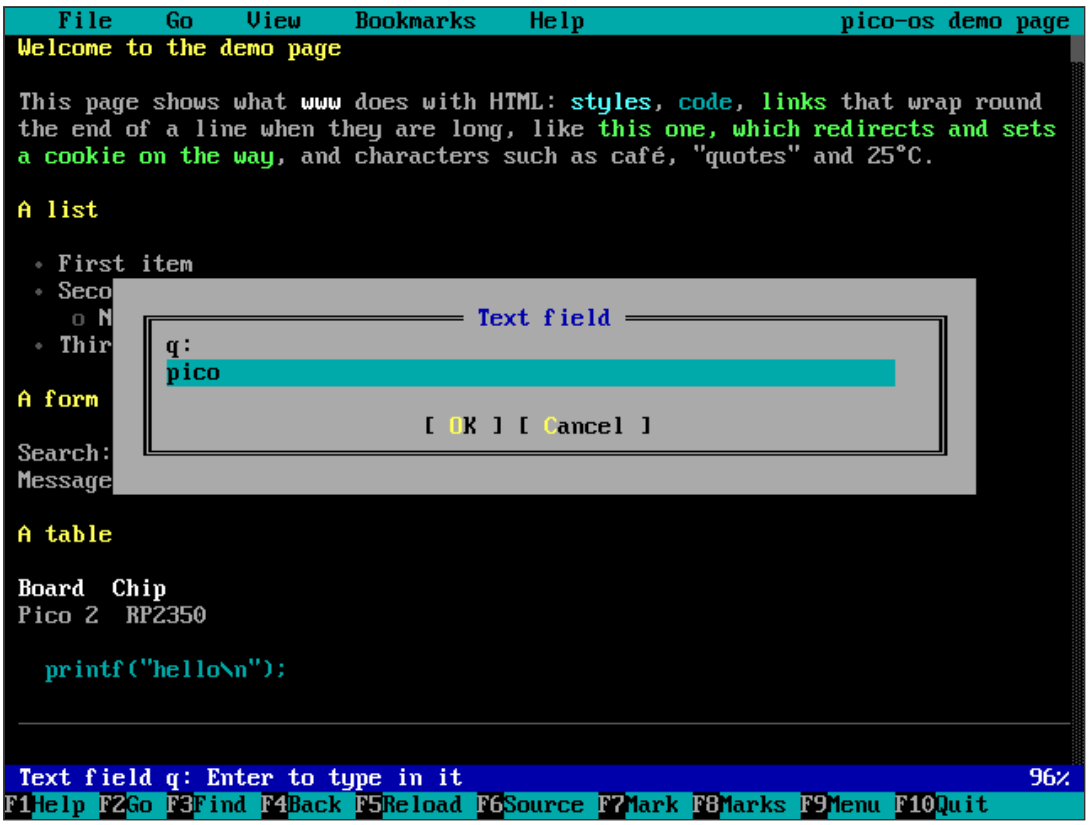
Keys

Key	
Up, Down, Tab	The previous or next link. When the next one isn't on the screen, the page scrolls half a screen.
Enter, Right	Follow the link, or work the form field (below).
Left, Backspace, F4	Back. Ctrl-F goes forward again.
PgDn, Space / PgUp, b	A screen down / up.
Home / End	The top / the end of the page.
g	Go to an address (or type words to search for them).
G, F2	Edit this page's address and go there.
o	Open a file on the drive.
/, F3	Find text on the page (any case); **n** finds it again (from the top after the end).
\, F6	The page's source (the HTML as it came), and back.
r, F5	Load the page again.
m	The start page.
a, F7 / v, F8	Bookmark this page / show the bookmarks.
d	Download the chosen link's file to the drive.
s / p	Save the page's source / save it as text (as shown).
=	About the page and the chosen link.
Ctrl-Y	The history: pick a page to go back (or forward) to.
h, F1	Help (the keys).
q, F10	Quit.



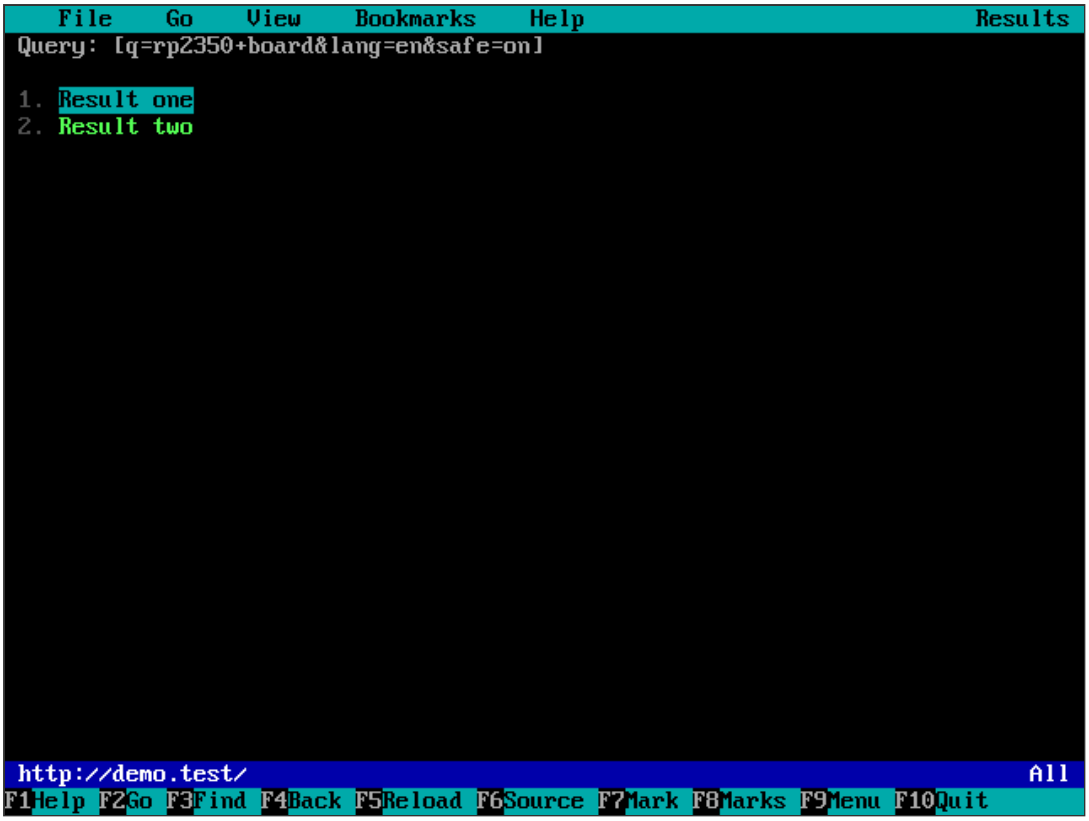
The mouse: click a link to follow it, right-click to go back; the wheel scrolls; click the menu bar, the key bar or the scroll bar.

Forms



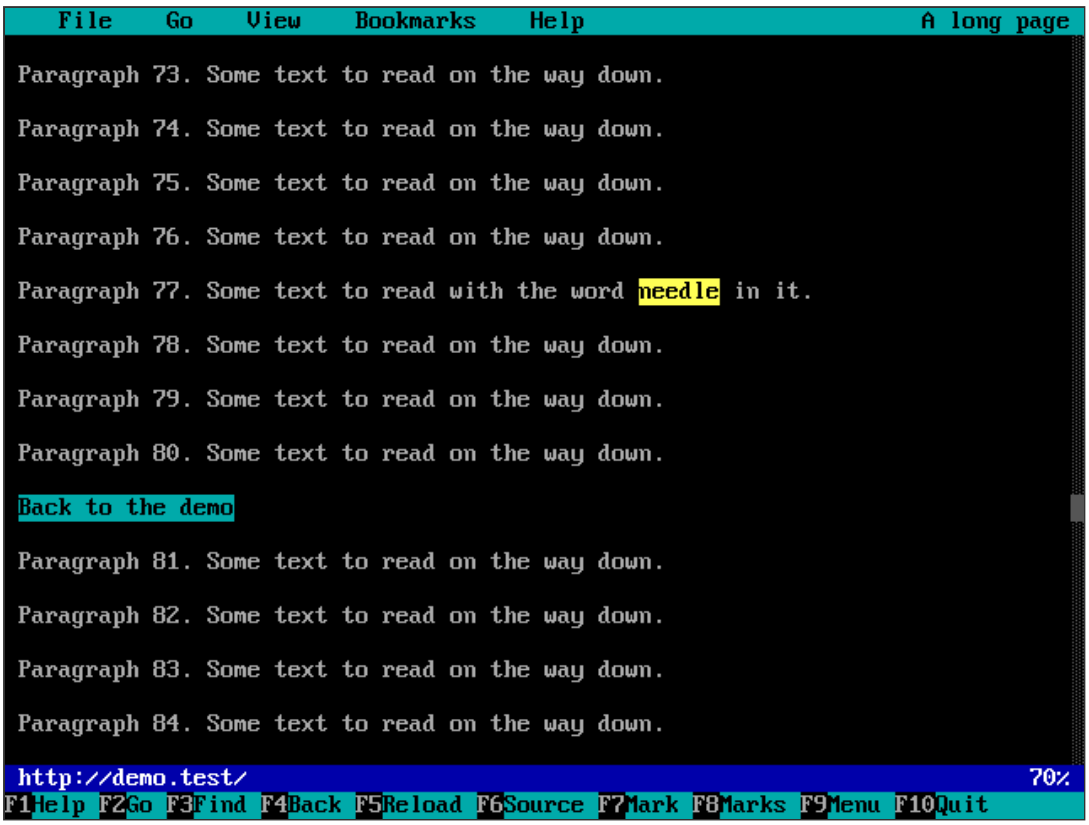
Search boxes and simple forms work. Choose a field with Up and Down, then Enter:

- **A text field** (or a password, or a text area) opens a box to type in (Ctrl-U clears it). In a form with only one text field, such as a search box, Enter in the box sends the form at once.
- **A checkbox** is ticked or cleared; **a radio button** is chosen (the others of its group are cleared).
- **A list** opens the choices to pick from.
- **A button** (`[Search]`, `[Go]`) sends the form, by GET or POST as the page asks.



Cookies a site sets are kept (in PSRAM) until the browser closes, so logins and sites that need a cookie work while it is open.

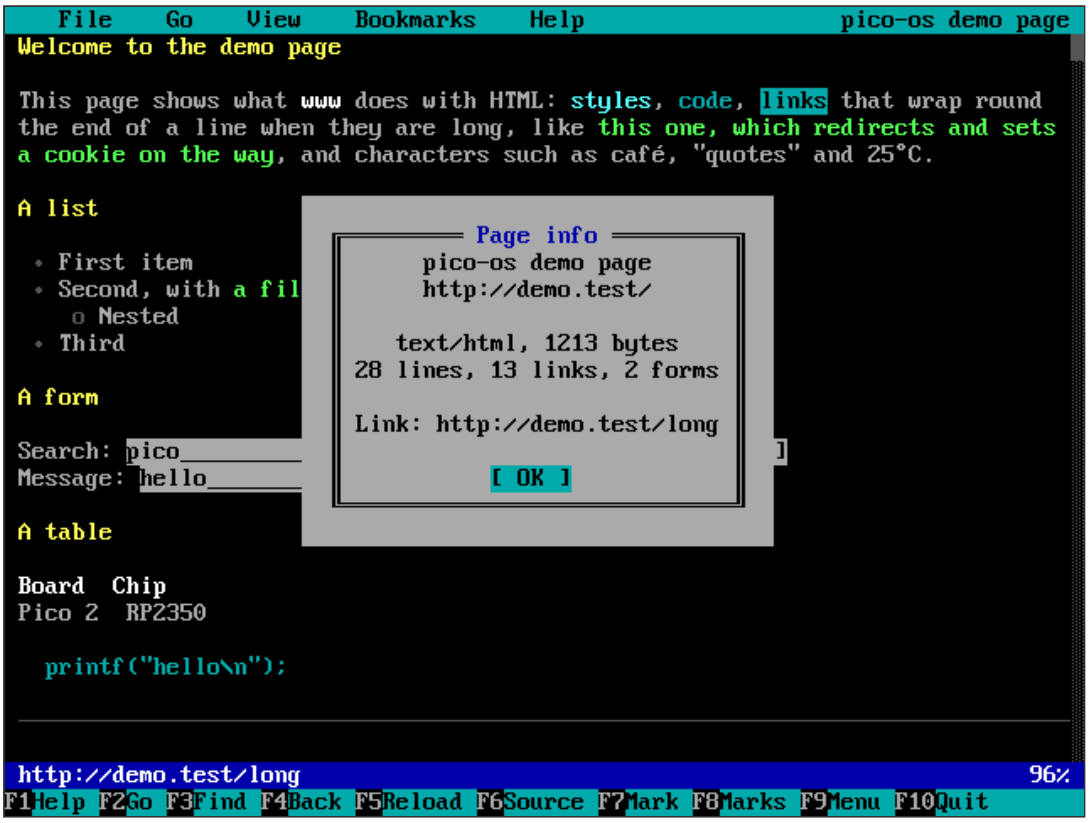
Finding text



`/` asks for the text; the page scrolls to it and it is shown in yellow (and a link on that line is chosen). `n` finds the next one.

Bookmarks, saving and downloading

- **a** adds the page to `/WWW/BOOKMARK.HTM` (an HTML page of links: edit it with `edit` if you like); **v** opens it.
- **d** downloads the chosen link to the drive (it asks for the name). Following a link to something that isn't a page (a ZIP, a picture, a program) asks the same. Files up to 4 MB go through PSRAM; for bigger ones use `wget`.
- **s** saves the page's HTML, **p** the page as text (CRLF lines, as on the screen).
- `/WWW/HOME.HTM` on the drive replaces the start page: any HTML page, for example your bookmarks copied there.



What it does and doesn't do

- HTML is read in one pass: paragraphs, headings, lists (numbered and nested), definition lists, quotes, `<pre>` text, rules, tables (each row on a line, the cells side by side), links, anchors (`#name`), frames and meta refreshes (as links), forms, entities and UTF-8 or Latin-1 text. Characters the screen's font (code page 437) lacks are shown as the nearest one (accents dropped, curly quotes straight) or `?`.
- Not done: pictures, JavaScript, CSS, layout tables side by side, text areas of more than one line, file uploads, compressed replies (the browser doesn't ask for them), HTTP/2.
- Pages up to 4 MB are loaded; bigger ones are cut there.
- https: the certificate is checked against `CACERT.PEM` (see the *Network Tools* manual). If it can't be checked, the browser asks whether to load the page anyway (Yes, No, or Always for this session); View > Check certificates turns checking off and on.
- It identifies itself as lynx, so sites send their simple pages.

Memory

Everything the browser keeps is in PSRAM, and only while it is open: the pages it has shown (up to 3 MB of them, so Back doesn't load them again; older ones are loaded again when needed), each page laid out, the history (40 pages), the cookies. While a page loads, the network uses the same SRAM as `wget` (a 16 KB receive buffer, and about 40 KB more for https).

Testing on a PC

`tools/wwwtest/uitest.sh /absolute/out/folder` builds the browser for a PC with its real screen, menus, dialogs and fetcher against a pretend web server, drives it with keys and the mouse (pages, links, a redirect that sets a cookie, the forms by GET and POST, find, the source, a download, bookmarks, the history, help), checks what the screen shows, and writes the screenshots in this manual (`www_*.png`). It also lays out thousands of random and cut-off pages and checks URLs are resolved as browsers do, all under AddressSanitizer.